COMPUTER ARCHITECTURE & ORGANIZATION

OVERVIEW OF COMPUTER ARCHITECTURE & ORGANIZATION



Prepared By Nigatu A.

1. INTRODUCTION

- **Digital Computer:**
  - It is a digital system that performs various computational tasks
  - Digital computers use the binary number system, which has two *digits:* 0 and 1.
  - Information is represented in digital computers in groups of bits.
  - By using various *coding* techniques, group of bits can be made to represent not only binary numbers but also other discrete symbols, such as decimal *digits or* letters of the alphabet
  - By judicious use of binary arrangements and by using various coding techniques, the groups of bits are used to develop complete sets of instructions for performing various types of computations.

Computer system

- A Computer system is sometimes subdivided into two functional entities: **Hardware and Software**
- **The Hardware:**
  - The hardware of the computer consists of all the electronic components and electromechanical devices that comprise the physical entity of the device.
- **Software:**
  - Computer software consists of the set of instructions and related data that the computer manipulates to perform various data-processing tasks and telling a computer what to do and how to do it.
  - A sequence of instruction for the computer is called a program.
  - So that, Software is a set of *programs, procedures, algorithms and its documentation concerned with the operation of a data processing* system.

Classes of Software

- Computer systems divide software systems into two major classes: *System software and Application software.*
- **System software:**
  - System software provides the basic functions for computer usage and helps to run the computer hardware and system.
  - ***For example:*** the compiler that translates the high-level language program to machine language is a system program.
  - The System software is indispensable(Absolutely necessary) part of a total computer system.
  - Its function is to compensate for the differences that exist between user needs and the capability of the hardware.
  - It includes : Device drivers, Operating systems, compilers, Assemblers

Cont'd

- **Application software:**
  - Application software is developed to perform in any task that benefits from computation and for the purpose *of solving particular problems.*
  - *Example: a high-level language program written by a user to solve particular data-processing needs is an application program.*
  - It is a set of programs that allows the computer to perform a specific data processing job for the user.
  - This category includes: Media player software, Internet browsers, Image editing, Mathematical software, Simulation software, Spreadsheets, Video editing software, Word processing, AutoCAD.

Parts of Computer hardware

- The hardware *of the computer is usually divided into three major parts:* as shown in Fig.1.1.
  - ➢The central processing *unit (CPU)*
  - ➢The memory
  - ➢The input and output processor(IOP)

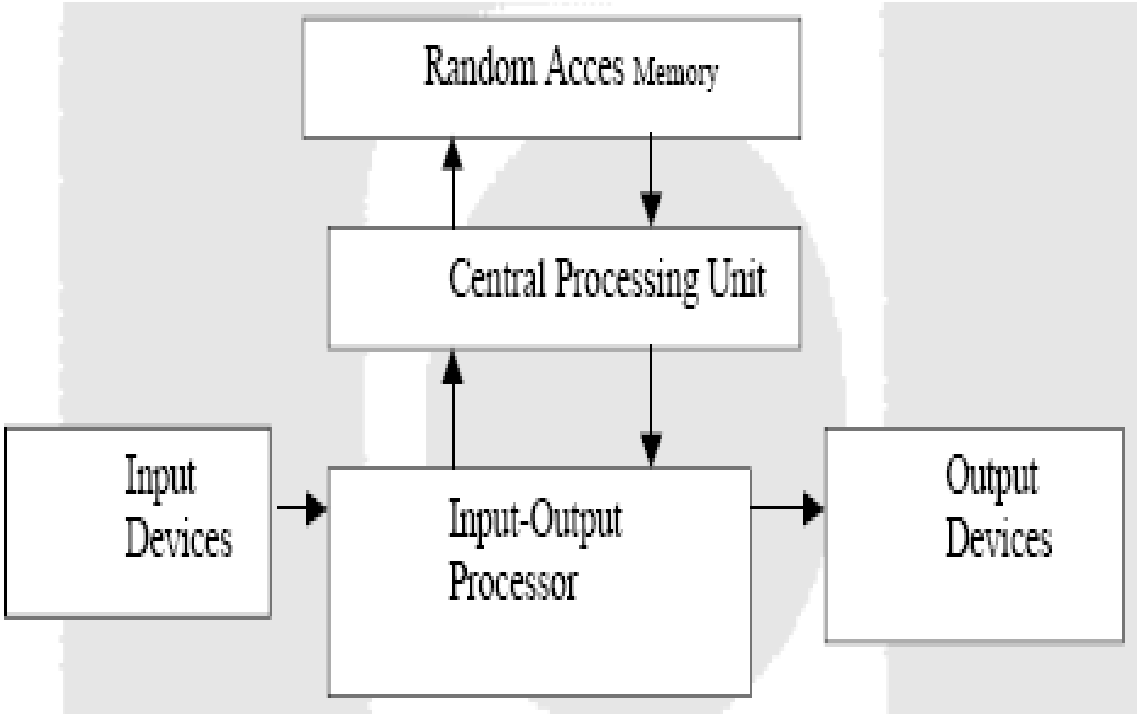


Fig.1.1. Block diagram of a digital computer

Cont'd

- ***The central processing unit (CPU) :***
  - ***Contains an arithmetic*** **and logic unit for manipulating data, a number of registers for storing data, and control circuits for fetching and executing instructions.**
- ***The memory of a computer :***
  - **Contains storage for instructions and data.**
  - **It is called a random access memory (RAM) because the CPU can access any location in memory at random and retrieve the binary information within a fixed interval time.**
- ***The input and output processor (IOP) :***
  - **Contains, electronic circuits for communicating and controlling *the transfer of information between the computer and the* outside world.**
  - **The input and output devices connected to the computer include: keyboards, printers, terminals, and other communication devices.**

Computer Organization, Computer Design, Computer Architecture

- When dealing with computer hardware it is customary to distinguish between what is referred to as:
  - Computer Architecture
  - Computer Organization and
  - Computer Design

cont'd

- **Computer Architecture:**
  - It is concerned with the specifications of the various functional modules, such as processors and memories and structuring them together into a computer system.
  - It uses higher-level building blocks to optimize the overall system performance
  - A computer architect, however, is not concerned with the internal details of the components(e.g. ALU).
  - Architecture is those attributes visible to the programmer includes: Instruction set, number of bits used for data representation, I/O mechanisms, addressing techniques.
  - For example: It is an architectural design issues whether the computer system will have a multiply instructions.

Cont'd

- **Computer Organization:**
  - It is concerned with the way the hardware computer operate and the way they are connected together to form the computer s/m.
  - The various components are assumed to be in place and the task is to investigate the organizational structure to verify that the computer parts operate.
  - It refers to the operational units and their interconnections that realize the architectural specifications.
  - Organizational attributes includes those hardware details transparent to the programmer such as control signals, interfaces between the computer and peripherals, and memory technology.
  - For example: It is an organizational issues whether the multiply instruction will be implemented by special multiply unit or by a mechanism that makes repeated use of the add unit of the system.
  - Organizational decision may be based on speed, costs or physical sizes.

Cont'd

- **Computer Design:**
  - It is concerned with the hardware design of the computer.
  - Once the computer specification is formulated, it is the task of the designer to develop hardware for the system.
  - It is also concerned with the determination of what hardware should be used and how the parts should be connected.
  - This aspect of computer hardware is sometimes referred to as computer implementation.
  - Implementers are responsible for implementing the designs produced by computer architects.
  - This group works at the digital logic level. At this level, logic gates and other hardware circuits are used to implement the various functional units.

Cont'd

- **The difference between computer architecture and computer organization:**

Computer Architecture	Computer Organization
1. It includes emphasis on logical design, computer design and the system design.	It includes emphasis on the system components, circuit design, logical design, structure of instructions, computer arithmetic, processor control, assembly language programming and methods of performance enhancement.
2. It is concerned with the structure and behaviour of computer as seen by user.	Computer organization is concerned with the way the hardware components operate and the way they are connected together to form the computer system.